

4. [Maximum mark: 6]

A particle moves in a straight line such that its velocity, $v \text{ m s}^{-1}$, at time t seconds is given by $v(t) = 1 + e^{-t} - e^{-\sin 2t}$ for $0 \leq t \leq 2$.

- (a) Find the velocity of the particle at $t = 2$. [1]
- (b) Find the maximum velocity of the particle. [2]
- (c) Find the acceleration of the particle at the instant it changes direction. [3]

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